Epistemic modality and the true nature of eavesdropping judgements

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Introduction

The truth conditions of epistemic modal utterances like 'Joe might be in Boston' are heavily disputed. **Contextualists** argue they are true iff the prejacent is compatible with the utterer's information (von Fintel & Gillies 2008; Dowell 2017) **Relativists** counter-argue the relevant information is the assessor's (Egan 2007, MacFarlane 2011). Experiments on **eavesdropping judgments of might-statements** are central to this debate (Knobe & Yalcin 2014; Beddor & Egan 2018; Khoo & Phillips 2019; Phillips & Mandelkern, forth.). Here, we **focus on Knobe & Yalcin (2014; KY)'s exp. 4 and Phillips & Mandelkern (PM)'s replication**.

KY (Exp 4) and PM (Replication)

In this study, participants read a context where a character utters 'might p' but later discovers that 'p' is false (eavesdropping context).

An eavesdropping scenario: *Sally and George are talking about whether Joe is in Boston. Sally carefully considers all the information she has available and concludes that there is no way to know for sure. Sally says: "Joe might be in Boston." Just then, George gets an email from Joe. The email says that Joe is in Berkeley. So George says: "No, he isn't in Boston. He is in Berkeley."*

Judgement: *Please tell us whether you agree or disagree with the following statement: What Sally said is false.*

Scale: 1 ("completely disagree") to 7 ("completely agree").

Results (N = 42): a mean rating slightly above 3

KY & PM's interpretation: English speakers don't tend to judge 'might p' false in eavesdropping scenarios, falsifying the relativist prediction (implicitly supporting contextualism)

Discussion: the authors interpret their results as providing evidence against the relativist prediction that English speakers tend to judge 'might p' false if their evidence denies 'p' no matter the utterer's evidence. However, their findings equally undermine the contextualist prediction that English speakers tend to judge 'might p' true based on the utterer's information. **Here, we show in two experiments that English speakers behave according to neither theory, answering at chance in binary truth-value judgments of justified 'might p' utterances across multiple eavesdropping contexts and question conditions, and propose three potential explanations.**

Exp 1: is 'might p' true or false in eavesdropping?

Study goals: 1. Test if the distribution of judgments in KY Exp 4 was due to task effects or problems with the scenario, and **2.** Assess whether binary judgments support contextualism.

Methodology: Adopting some methods of Doran et al. 2012, we modified both the scenario and the question of KY Exp 4:

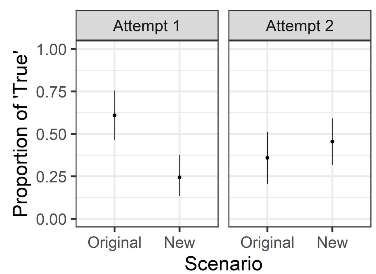
- <u>Scenario:</u> Some issues with KY scenario: a) participants are not told any positive evidence for Sally's claim; b) participants have to assume George's perspective; c) there is short time gap between when George hears the statement and when he receives the news contradicting 'p'
Therefore, we compared the original scenario to a modified version where: a) Sally's evidence supporting her 'might p' claim is made explicit; b) Sally is talking directly to participant who already knows 'p' is false

New scenario: Imagine that you are sitting in a café in Berkeley with a friend of yours named Sally, having a conversation about another friend, Joe. Sally says: "<u>Joe might be in Boston</u>, since he had a business meeting scheduled there this week." You know that Joe's meeting in Boston was cancelled and he is in Berkeley right now.

- <u>Question/Task:</u> Critical 'might p' statement was underlined. Participants were then asked 'Is the underlined statement true or false?'
(Rationale: mentioning the utterer as in 'What Sally said is false' may prompt some participants to interpret 'false' as 'infelicitous')
Participants chose from two response options: True False
(Rationale: We forced participants to take a stance and simplified the potentially unclear question phrasing from KY)

Predictions: Contextualism: proportion of 'True' > chance
Relativism: proportion of 'True' < chance

Results: (We ran the experiment twice; N=50 per condition)



No significant difference from chance except in Attempt 1 New condition, and participant behavior on the original KY is unstable across instances of the experiment! But even the new scenario is at chance.

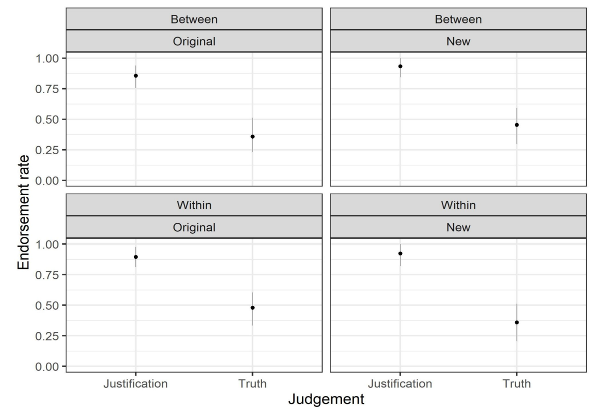
Discussion: English speakers tend towards chance behavior, supporting neither contextualism nor relativism. This behavior isn't attributable to potential confusion about the task or scenario present in KY. What underlies this divide? Are speakers split in two populations of contextualists and relativists? If not, are choices random or influenced by any experimental factors?

Exp 2: truth vs justification

Study goal: assess the contextualist prediction that 'might p' is true iff the utterer's evidence justifies her to make the claim by testing the relationship between justification and truth judgments

Methodology: Three factors in a 2X2X2 design
<u>a. Scenario:</u> As in Exp1 (*Original* vs *New*);
<u>b. Judgment:</u> *Justification*: Based on what she knows, is Sally justified to say the underlined statement? Justified Not justified
Truth: Is the underlined statement true or false? True False
<u>c. Design:</u> One judgment per participants (*Between*) vs both judgments with Justification before Truth (*Within*)

Results: (N=50 per condition)



Discussion: Participants converge in judging Sally justified to utter 'might p', yet divided in its truth judgment *even when they make both judgements together*, against contextualism. Truth judgments staying at chance is also against relativism.

Conclusion

We showed that, although English speakers **converge in judging Sally justified** in uttering 'might p' in a revised eavesdropping scenario, they **split in their truth-value judgment** of the same statement, supporting **neither contextualism nor relativism**.

In order to use eavesdropping data for discriminating between contextualism and relativism, it is necessary to understand what underlies the split. We propose some possible explanations for future work to test:
1. Different populations disagree on the truth-conditions of 'might',
2. Different populations disagree on the meaning of 'true/false' judgments 3. Judgments are noisy because the primary meaning contribution of 'might' is not truth-conditional, and so the 'true/false' judgments are inapplicable to 'might p' utterances (and participants thus choose randomly)